Student Performance Analysis and Prediction using AI

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Domain: Data Analytics with AI

1. Introduction

This project demonstrates an end-to-end data analytics workflow for studying student performance using a synthetic dataset and machine-learning techniques.

2. Objectives

• Data preparation and cleaning
• Exploratory data analysis and visualization
• Final-score prediction
• At-risk student classification
• Student clustering
• AI-driven insights

3. Dataset

The notebook generates 500 synthetic student records with study hours, attendance, assignments completed, previous score, sleep hours, internet access, final score and an at-risk indicator.

4. Models

Linear Regression is used for final-score prediction. Logistic Regression and Decision Tree are used for at-risk classification. K-Means is used for unsupervised student clustering.

5. Insights

The workflow illustrates how study habits, attendance and previous performance can be examined with predictive models. Clustering provides broad performance patterns.

6. Limitations

The data are synthetic and should not be treated as real student records or a validated educational assessment. Real deployment would require privacy, fairness, validation and monitoring.

7. Conclusion

The project presents a complete analytics pipeline from data preparation and visualization to machine-learning-based insights.